# Test Plan and QA Strategy

Hasnahena Platform

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|---|---|
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*Quality assurance strategy*
*for financial integrity and security*

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# Document Control

| Item | Details |
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**Revision History**

| Ver. | Date | Author | Changes |
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| 1.0 | March 22, 2026 | Md Saif Ahmed | Initial release |

# 1 Purpose

This document defines the verification and validation strategy for the Hasnahena platform. The test plan is designed to ensure:

- **Financial Integrity:** All monetary calculations and ledger operations are accurate
- **Role Security:** RBAC enforcement prevents unauthorized actions
- **Workflow Correctness:** Request → Process → Confirm lifecycle works correctly
- **Compliance:** Shariah compliance and audit trail requirements are met

# 2 Test Scope

## 2.1 In Scope

- Authentication and authorization mechanisms
- Registration and approval workflows
- Deposit, withdrawal, loan, repayment, and donation flows
- Dispute handling and correction workflows
- Audit logging and immutability
- API contract validation
- Localization (English/Bengali)
- Dashboard functionality for all roles

## 2.2 Out of Scope (Phase 1)

- Mobile application testing
- Load/stress testing (deferred to Phase 2)
- Third-party payment gateway integration
- eKYC/NID API integration testing

# 3 Test Levels

| Level | Coverage | Tools |
|---|---|---|
| **Unit** | Validation logic, eligibility calculators, policy rules, settlement waterfall | Jest (frontend), JUnit (Java), Go testing |
| **Integration** | DB transaction atomicity, API handlers, audit append behavior, service communication | Testcontainers, Supertest |
| **End-to-End** | Request-process-confirm lifecycle across roles, notification hooks | Playwright, Cypress |

| Level | Coverage | Tools |
|---|---|---|
| **Security** | RBAC enforcement, session handling, brute-force protections, input validation | OWASP ZAP, custom scripts |
| **API Contract** | Endpoint validation, response structure, error codes | Postman, Pact |

## 4 Priority Test Scenarios

The following scenarios are considered **Critical** priority due to their impact on financial integrity:

1. **Incorrect Amount Dispute**
   - Worker processes wrong amount
   - Member raises dispute
   - Admin correction is fully traceable
2. **Loan Cap Enforcement**
   - Personal cap (5× deposit)
   - Fairness cap (2× pool/members)
   - Solvency cap (available funds)
   - Dynamic recalculation on deposit changes
3. **Default Settlement Waterfall**
   - Deduction from borrower deposit
   - Proportional deduction from guarantors
   - Charity fund write-off
   - Residual debt handling
4. **Withdrawal Blocking**
   - Active loan collateral ratio
   - Active guarantor obligations
   - Notice period enforcement
5. **Audit Log Immutability**
   - No modification of historical records
   - Correction operations create new entries
   - Admin cannot delete audit entries

# 5 Entry and Exit Criteria

## 5.1 Entry Criteria

Testing may begin when:

- Requirements baseline is approved and frozen for sprint
- API contracts are documented and versioned
- Database schema is deployed to test environment
- Test environment is provisioned and accessible
- Test data is prepared and loaded

## 5.2 Exit Criteria

Testing is complete when:

- **Zero Critical** defects remain open
- **Zero High** defects in core financial workflows
- All priority scenarios pass in staging environment
- Regression pass for RBAC and audit log invariants
- Test coverage meets minimum threshold (80% for core modules)
- Security scan shows no critical vulnerabilities

# 6 Defect Classification

## 6.1 Severity Levels

| Severity | Description & Examples |
|---|---|
| **Critical** | Financial inconsistency, unauthorized money movement, data corruption, security breach, system crash |
| **High** | Blocked core flow, role permission leakage, incorrect ledger projection, failed transaction that should succeed |
| **Medium** | Validation mismatch, UI state inconsistency, non-blocking workflow issue, minor calculation error |
| **Low** | Cosmetic issues, typos, non-critical localization defects, minor UX improvements |

## 6.2 Priority Matrix

| | **High Impact** | **Med Impact** | **Low Impact** |
|---|---|---|---|
| **High Frequency** | P1 - Immediate | P2 - Next Sprint | P3 - Backlog |
| **Med Frequency** | P2 - Next Sprint | P3 - Backlog | P4 - Future |
| **Low Frequency** | P3 - Backlog | P4 - Future | P5 - Optional |

# 7 Test Environment

## 7.1 Environment Specifications

| Component | Configuration |
|---|---|
| Frontend | Vercel Preview Deployments (per PR) |
| Backend | Fly.io staging instance |
| Database | Neon branch (isolated per feature) |
| Test Data | Seeded from fixtures, reset daily |

## 7.2 Test Data Requirements

- Minimum 50 member accounts across all states
- Loan scenarios covering all status values
- Transaction history spanning 12+ months
- Edge case data (zero balance, max limits, etc.)

# 8 Roles and Responsibilities

| Role | Responsibilities |
|---|---|
| QA Lead | Test planning, defect triage, sign-off |
| QA Engineer | Test case design, execution, automation |
| Developer | Unit tests, bug fixes, code review |
| Product Owner | Acceptance criteria, UAT sign-off |
| DevOps | Environment provisioning, CI/CD pipelines |

# 9 Approvals

**QA Lead**
Date: _____

**Technical Lead**
Date: _____